

CY 404: Fundamentals of Soft Matter Science and Applications

Course Code	CY 404
Course Name	Fundamentals of Soft Matter Science and Applications
L-T-P-C	3-0-0-3
Intended for	BS
Approval	54th BoA (May 2024)

Course Contents

- **Introduction to Soft Matter:** Definition and examples of soft matter, Self-assembly, and structures in soft matter, Intermolecular forces, and properties. (7 hours)
- **Liquid Crystals: Chemical and Physical Aspects:** Introduction and historical development of liquid crystals, Physical and structural properties, Anisotropy, Order parameter, Birefringence, Defect textures (demonstration of textures using polarizing microscopy), Alignment in liquid crystals, Classification of liquid crystals, Chirality in liquid crystals, Photoswitchable liquid crystals, Brief introduction to liquid crystal gels, colloids, nanoparticles, polymers and elastomers, Liquid crystals in biology. (15 hours)
- **Surfactants and Amphiphiles:** Micelles (cationic, anionic, and neutral, CMC determination and applications), Vesicles, Bilayers, Cylinders, Micro-emulsions, Ionic liquids, Lyotropic liquid crystals, Membranes, Self-assembly and their phase behavior. (7 hours)
- **Instrumentation and Characterization Techniques:** Polarized optical microscopy, X-ray scattering (XS): Small-angle (SAXS), Wide-angle (WAXS), Grazing incidence (GI-SAXS/WAXS), Differential Scanning Calorimetry (DSC), Rheology, Dynamic light scattering (DLS), Langmuir-Blodgett technique, Microfluidics. (6 hours)
- **Applications in Technology and Beyond:** Liquid crystals in displays, photonics, drug delivery, biological and chemical sensors, discotic LCs in organic opto-electronics. (7 hours)

Text Books

1. L. S. Hirst, **Fundamentals of soft matter science**, CRC press (2019).
2. J. W. Goodby, P. J. Collings, T. Kato, C. Tschierske, H. Gleeson, P. Raynes and V. Vill, **Handbook of liquid crystals**, (Vol. 1), John Wiley & Sons (2014).
3. P. J. Collings and J. W. Goodby, **Introduction to liquid crystals: chemistry and physics**, CRC Press (2019).
4. B. Bahadur, **Liquid crystal-applications and uses** (Volume 1), World scientific (1990).
5. R. Borsali and R. Pecora, **Soft-matter characterization**, Springer Science & Business Media (2008).

References

1. I. W. Hamley, **Introduction to soft matter: synthetic and biological self-assembling materials**, John Wiley & Sons (2007).
2. S. Kumar, **Chemistry of discotic liquid crystals: from monomers to polymers**, CRC press (2014).
3. A. A. Collyer, **Liquid crystal polymers: from structures to applications** (Vol. 1), Springer Science & Business Media (2012).
4. H. K. Bisoyi, and Q. Li, **Liquid crystals: versatile self-organized smart soft materials**, Chemical reviews, 2021, 122, 4887-4926.
5. S. W. Ula, N. A. Traugutt, R. H. Volpe, R. R. Patel, K. Yu, and C. M. Yakacki, **Liquid crystal elastomers: an introduction and review of emerging technologies**, Liquid Crystals Reviews, 2018, 6, 78-107.
6. A. M. Lowe, N. L. Abbott, **Liquid crystalline materials for biological applications**, Chemistry of Materials, 2012, 24, 746-58.
7. B. R. Kaafarani, **Discotic liquid crystals for opto-electronic applications**, Chemistry of Materials, 2011, 23, 378-396.
8. S. Setia, S. Sidiq, J. De, I. Pani and S. K. Pal, **Applications of liquid crystals in biosensing and organic light-emitting devices: future aspects**, Liquid Crystals, 2016, 43, 2009-2050.